Comparative Analysis of Solar Active Region Detection Techniques Using Aditya-L1 SUIT Data

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***Abstract*—This report presents a comparative study of segmentation techniques applied to solar imagery acquired by the Solar Ultraviolet Imaging Telescope (SUIT) onboard ISRO's Aditya-L1 spacecraft. Three sub-tasks are addressed: temporal evolution of active regions (Part 2A), intensity–magnetic flux correlation (Part 2B), and comparative segmentation methods (Part 2C). For Part 2C, Otsu thresholding and k-means clustering (k=3) are applied to a Level-1 SUIT NB03 image acquired on May 9, 2026 at 07:09:21 UTC. Otsu detected 61 blobs matching 7 unique NOAA active regions, while k-means detected 159 blobs (after minimum area filtering) matching 5 unique NOAA active regions. The results indicate that Otsu thresholding achieves superior recall of catalogued active regions under the present imaging conditions.**

I. Introduction

The Sun's active regions are sites of intense magnetic activity associated with solar flares, coronal mass ejections (CMEs), and space weather events. Accurate detection of these regions from solar imagery is a fundamental task in heliophysics. The Aditya-L1 mission, launched by ISRO in September 2023, carries the Solar Ultraviolet Imaging Telescope (SUIT) which observes the Sun in the near-ultraviolet, particularly in the Mg II k line at 2796 Å (NB03 filter). This report investigates three research questions using SUIT Level-1 data, each addressing a distinct aspect of active region analysis. Part 2A examines temporal evolution, Part 2B investigates intensity–magnetic flux correlation, and Part 2C performs a comparative evaluation of segmentation methods.

All analyses use a SUIT NB03 Level-1 FITS image dated May 9, 2026 at 07:09:21 UTC as the primary dataset. The image is off-centered, introducing distortions in the western and southern hemispheres, which are accounted for in preprocessing.

II. Part 2A: Temporal Evolution of Active Regions

A. Research Question

Using a sequence of Aditya-L1 images over several days, can you quantify how the area of a specific active region changes with time?

B. Dataset

Six Level-1 SUIT NB03 FITS files (Mg II k, 2796 Å) were obtained from ISRO's PRADAN portal, one per day from May 7–12, 2026, each acquired at approximately 07:09 UTC. Active region AR14436 was selected as the tracking target because of it being a major solar event owing to its production of an M5.7 flare and CME on May 10, 2026, which makes it an event to be watched and analysed.

C. Methodology

The pipeline that we had for Part 1 of the project was used here with the necessary modifications applied as per the need of this part. The pipeline was extended into a loop structure that processed all six FITS in date-wise sequence. For all the files, the same pre-processing steps were applied: flat-field correction, percentile normalization, and circular disk masking. For circular disk masking, the radius was further tightened from 775 to 680 pixels, in order to minimize the off-disk artifact from all the six images. This was then followed by Otsu thresholding which was applied to the normalized array having intensity values greater than 0.1. After this morphological operations were employed to remove the solar limb and the on-disk vertical and horizontal patches.

Heliophysics Event Knowledgebase (HEK) was then used to retrieve the heliographic position of AR14436. All the blobs'- which were detected as AR14436-area was then summed up to give a single value per day. This was employed as AR14436 was fragmented in our pipeline due to Otsu's limitation as discussed earlier in Part 1 of the project.

D. Results

TABLE I
AR14436 Detected Area Across Six Days

Date	AR14436 Area (pixels)
May 7, 2026	Not detected
May 8, 2026	Not detected
May 9, 2026	Not detected
May 10, 2026	4758
May 11, 2026	6047
May 12, 2026	6483

AR14436 was undetected on May 7–9, because at this time duration it was present near the western limb of the sun and this western limb (longitude of approximately $-81°$), due to further tightening of radius (from 775 to 680), was partially cut from the masked image. However, it was detected on 10 May which coincides with the M5.7 flare and CME event. It then, showed a monotonically increasing area till May 12. The detailed reason for this monotonically increasing behaviour of the trend is discussed in the next section. The temporal evolution is shown in Fig.1 .

E. Discussion

The monotonically increasing area trend ($4758 \rightarrow 6047 \rightarrow 6483$ pixels) is not indicative of physical growth. AR14436 was

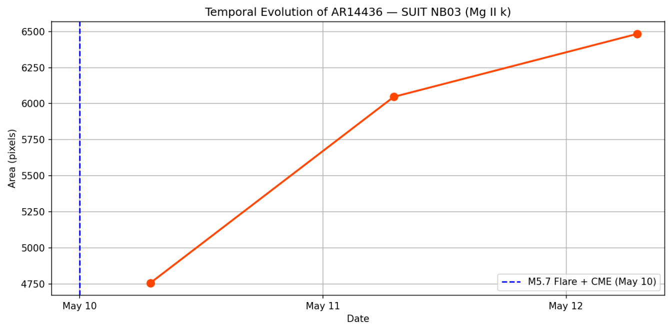

Fig. 1. Temporal evolution of AR14436 pixel area from May 7–12, 2026. The vertical dashed line marks the M5.7 flare and CME event on May 10. Non-detections on May 7–9 reflect the region's proximity to the western limb.

still emerging from behind the western limb on May 10. The increasing pixel count on other days is attributable to reduction of geometric foreshortening because the region rotates toward disk center. The other reason for lesser area of May 10 region is because the region AR14436 was on the region near western limb of the sun. Some of the area of the region may have been masked by the radius tightening (from 775 to 680). Due to this masking of some portion, its area is smaller than on the subsequent days. The other constraint that was that the analysis was limited to single image per day.

III. Part 2B: Intensity–Magnetic Flux Correlation

A. Research Question

Is there a measurable correlation between the brightness intensity of active regions in Aditya-L1 imagery and their associated magnetic flux concentrations?

B. Instrumentation Constraints

As per the project document, it was asked to preprocess images, extract intensity values, and overlay with magnetogram data. The data was subject to availability. This analysis was, however, not carried out due to a fundamental constraint. Aditya-L1 carries a magnetometer (MAG) instrument but, the issue is that the magnetogram measures the interplanetary magnetic field at the sun–earth L1 point as a scalar/vector time series at a single location. It doesn't produce two-dimensional maps of solar surface magnetic field which are required for this particular part of the project.

The planned methodology for Part 2B was to overlay co-registered SDO/HMI line-of-sight magnetograms onto SUIT NB03 intensity images and quantify the spatial correlation between UV brightness and photospheric magnetic flux density. However, this analysis was not carried out due to a fundamental instrumentation constraint.

The suitable data product for intensity–magnetic flux correlation studies is the HMI line-of-sight magnetogram from NASA's Solar Dynamics Observatory, providing full-disk photospheric field maps co-temporal with SUIT observations. A natural extension of this work and a recommended avenue for further analysis is the combination of SDO/HMI magnetograms with SUIT imagery by means of coordinate-aligned reprojection.

IV. Part 2C: Comparative Segmentation Methods

A. Research Question

How do different segmentation techniques — specifically Otsu thresholding and k-means clustering — affect the detection accuracy of active regions in Aditya-L1 SUIT data?

B. Dataset

The dataset used is a Level-1 SUIT NB03 FITS file (Mg II k line, 279.6 nm) acquired on May 9, 2026 at 07:09:21 UTC, sourced from PRADAN portal. The image has a spatial resolution of 2048 $\times$ 2048 pixels. This date was selected as a substitute for the originally specified date (31 December 2023), as the specified date's data is unavailable on PRADAN portal because Aditya L1 had not entered its halo orbit around Sun-Earth L1 point.

C. Preprocessing

To ensure a valid comparison, we applied both the methods to the same preprocessed array (`arr_2d_masked`). The preprocessing process consists of:

1) Flat-field correction using uniform filter of size 100 pixels
2) Percentile normalization (1st–99th percentile) which was then clipped to [0, 1]
3) Circular disk masking (radius = 775 pixels, center = (1320, 650) after performing multiple iterations) to minimize off-disk background as much as possible.

The resulting array `arr_2d_masked` was the common input to both segmentation methods. But, as morphological operations were also performed before Otsu thresholding and contour overlay was done it had to be done for K-means clustering too. Hence, morphological operations succeeded preprocessing.

D. Method 1: Otsu Thresholding

Otsu's algorithm works best in bimodal distribution. It automatically assigns a threshold value to the given data in the valley lying between two peaks. This threshold divides our data into non-active regions and active-regions. In our case, the threshold was found to be 0.5341 which was applied to solar disk pixels having normalized intensity greater than 0.1. This was done in order to prevent the disturbance created by off-disk background that was present in the masked image.

After thresholding, morphological operations were employed to remove solar limb, on-disk vertical and horizontal patches and to minimize the off-disk distortion. This was done because they in reality are not part of active regions but, algorithmically they were classified as active by Otsu because they had intensity values higher than 0.5341.

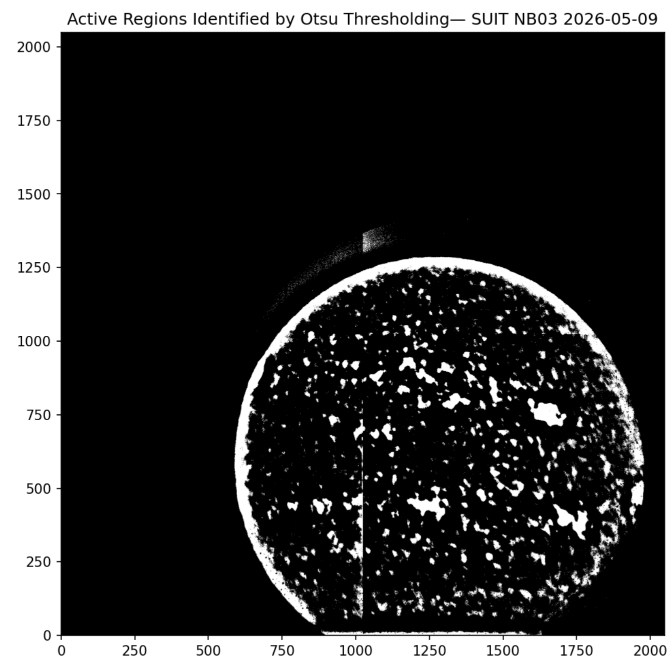

Fig. 2. Active regions identified by Otsu

E. Method 2: K-Means Clustering

K-means clustering was applied with k=3 clusters to the flattened `arr_2d_masked` array, reshaped to (4,194,304 × 1). The value k=3 is taken because of three physically distinct intensity groups present in the image. They are- the off-disk black background, the quiet sun and the brightest regions (active regions + solar limb + off-disk distortions). The algorithm was initialized using k-means++ with n_init=10 to minimize sensitivity to initialization. The algorithm converged in 9 iterations, producing three cluster centers:

TABLE II
K-MEANS CLUSTER CENTERS AND THEIR PHYSICAL INTERPRETATION

Cluster	Center Intensity	Physical Region
0	0.00	Off-disk background
1	0.47	Quiet Sun
2	0.63	Bright features (Active regions + solar limb + off-disk distortions)

The bright cluster (label = 2) was extracted as a binary mask. The same morphological operations which were performed to Otsu were also applied here: limb removal, on-disk horizontal and vertical patch removal and off-disk distortion minimization. A minimum threshold of 100 was applied to filter out the noise from K-means pixel-level clustering.

F. Results

Figure 2 shows the three-cluster segmentation map produced by k-means, with off-disk background (orange), quiet Sun (red), and bright features (grey) clearly distinguished. Figure 3 presents a side-by-side contour overlay comparison of both methods on the SUIT false-color image.

TABLE III
COMPARATIVE RESULTS: OTSU VS K-MEANS

Metric	Otsu	K-Means
Effective threshold	0.5341	0.63
Total blobs detected	61	159
Min area filter	None	100 px
Unique NOAA ARs	7	5

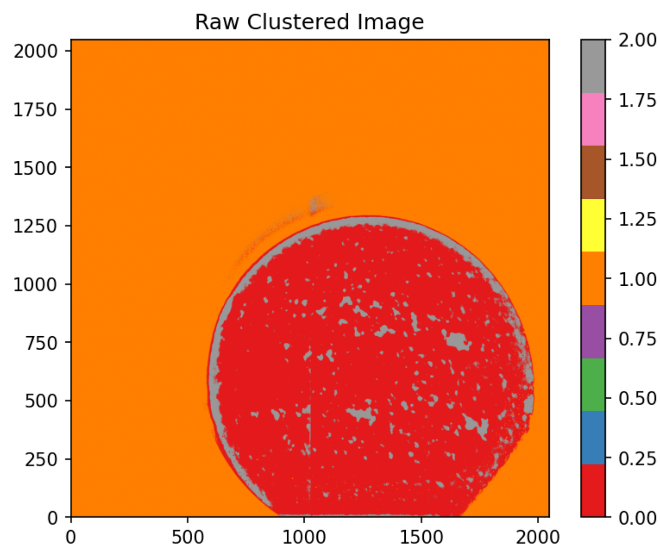

Fig. 3. K-means Clustered Image

G. Discussion

Both methods were applied to the same preprocessed array, in order to eliminate any difference that can arise due to different ways of preprocessing. It ensures that differences occur to algorithmic behaviour only.

K-means detected 159 blobs compared to Otsu's 61, yet matched 2 less unique NOAA active regions (5 vs 7). Two factors explain this:

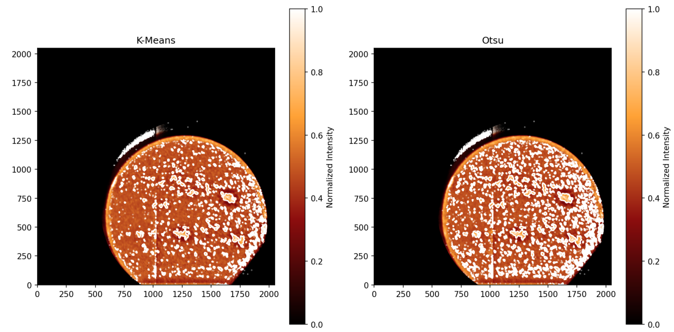

Fig. 4. Otsu vs K-means

Higher effective threshold: The k-means bright cluster center (0.63) is significantly higher than Otsu's threshold (0.5341). This means that the weaker active regions that exist in between this range 0.5341–0.63 are assumed to be quiet sun regions by K-means.

Greater pixel-level fragmentation: K-means algorithm works by assigning each pixel independently to the nearest cluster center without keeping spatial connectivity constraints in consideration. Due to this many isolated pixel groups are produced which are then neglected once we apply the minimum area filter of 100 pixels.

These two reasons explain why the number of detected active region blobs is higher in K-means clustering and its failure to match more or at least equal unique NOAA active regions.

Visually, both the methods are in strong agreement in the eastern hemisphere. However, significant differences can be observed in the western hemisphere of the sun, here weaker active regions are captured by Otsu (owing to lower threshold value) but, K-means fails to identify them as active (because of higher threshold value).

V. Discussion & Comparison Across Sub-Tasks

A unified discussion comparing findings across Parts 2A, 2B, and 2C will be presented here upon completion of all sub-tasks.

VI. Conclusion

For Part 2C of the project, we can conclude that Otsu thresholding provided us with better results than K-means clustering. The former matched 7 unique NOAA regions in comparison to the latter's 5. However, K-means clustering provides us the advantage of separating the data into more than 2 (in our case 3) parts. This is highly useful for visualization purposes in our work as it separates bright regions, quiet sun and off-disk black background which is a physically meaningful result. Both the methods provide their own advantage, while Otsu can be preferred for validating maximum regions, K-means offers us a highly visually interpretable cluster structure.

Conclusions for Parts 2A and 2B will be added upon their completion.

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